

# TrES-3b

R-2756

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_140617 · created 2026-09-02T23:58:35+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2456825.7134 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.155 +/- 0.046                 |
| Transit depth (Rp/Rs)^2                           | 2.39 +/- 1.42 [%]               |
| Orbital Inclination (inc)                         | 84.71 +/- 2.36                  |
| Airmass coefficient 1 (a1)                        | 0.995 +/- 0.05                  |
| Airmass coefficient 2 (a2)                        | 0.014 +/- 0.034                 |
| Scatter in the residuals of the lightcurve fit is | 4.98 %                          |
| Best Comparison Star                              | #2 - [522, 118]                 |
| Optimal Aperture                                  | 4.23                            |
| Optimal Annulus                                   | 14.77                           |
| Transit Duration (day)                            | 0.069 +/- 0.014                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.155 ± 0.046 vs 0.16309 (deviation 0.08σ) |
| Duration fit vs published | 0.069 d vs 0.0567 d (deviation 0.38σ)      |
| Mid-time O–C              | -1.0 min                                   |
| Depth SNR                 | 1.5                                        |
| Residual scatter          | 4.98 %                                     |

### Light curve

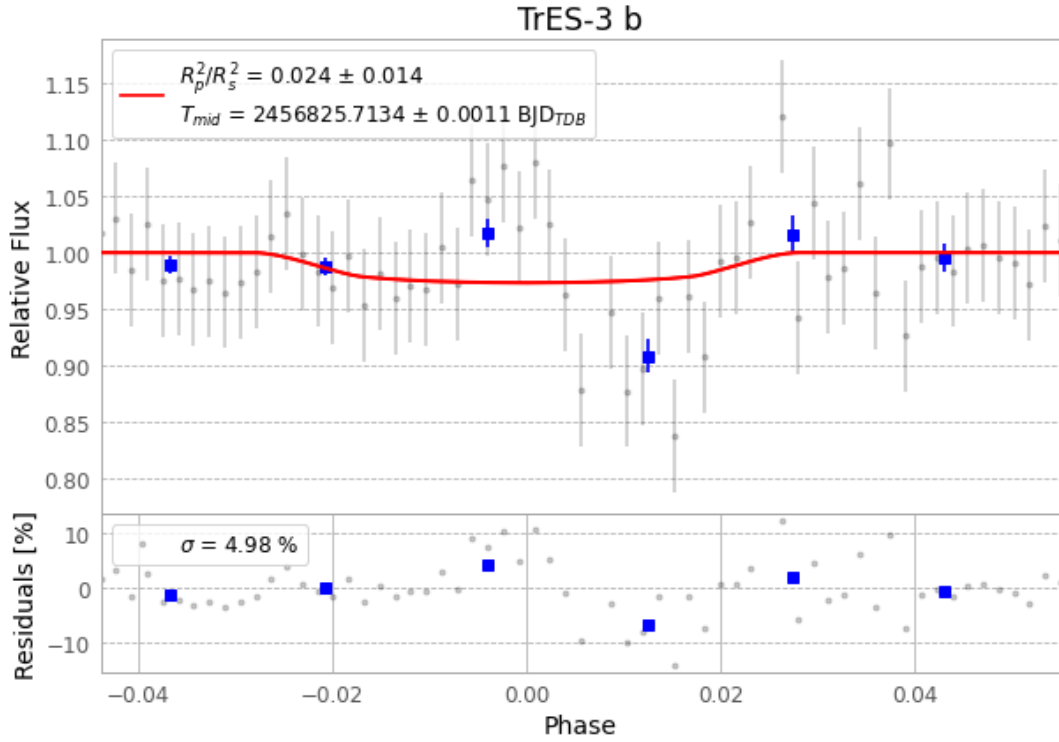


Figure 1 — FinalLightCurve\_TrES-3 b\_2014-06-16.png (SHA-256 8467f3b5c931fde4...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 228], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 bf2fede1c198edc3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

63 FITS frames (41 MB), TRES-3140617033912.FITS → TRES-3140617064513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140617.log (5594502bc92f...), FinalLightCurve\_TrES-3 b\_2014-06-16.png (8467f3b5c931...), FinalLightCurve\_TrES-3 b\_2014-06-16.csv (90cc3e2ad67a...)

## Compute resources

Wall time 127.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:58Z.